Karatina University

Department of Informatics and Innovation

Department:	Informatics and Innovation
Programme:	BSc – Computer Science (P101)
	Bachelor of Information Technology (P100)
	Bachelor of Science with Education (P106)
	B.Sc Information Science (P108)
	Bachelor of Science (E103)
Course Code:	COM 225
Course Title:	OPEN SOURCE SYSTEMS
Credit Hours:	3 hrs
Academic year:	2025/2026
Time:	1.00 PM-4.00PM, Thursday
Lecturer:	Dr. Gilbert Mugeni (PhD)
Contacts:	email: **gbmugeni@karu.ac.ke**
	Cell phone: +254 722 803 894
Office:	School of Computing and Informatics

Days of the week: Wednesday, Friday

COM 225 OPEN SOURCE SYSTEMS

Course description

The **Open Source Systems** undergraduate course introduces students to the principles, technologies, and collaborative practices that define open source software development. The course examines the historical foundations of the movement, including the influence of the Free Software Foundation and the Open Source Initiative in promoting software freedom and open licensing standards. Students explore key concepts such as open source licensing, community governance, distributed development, and sustainability models. Practical components provide hands-on experience with tools like Git and the Linux environment, enabling learners to participate in collaborative coding, version control, and system configuration. The course also analyzes the role of open source technologies in enterprise systems, cloud computing, cybersecurity, and emerging digital innovations. By the end of the course, students will be equipped to evaluate, deploy, and contribute to open source solutions in both organizational and community-driven contexts.

Course objective:

The objective of the **Open Source Systems** undergraduate course is to equip students with foundational knowledge and practical skills in open source software development, deployment, and governance. The course aims to develop learners' understanding of open licensing frameworks promoted by organizations such as the Open Source Initiative and the collaborative development culture surrounding platforms like Linux. Students will gain hands-on experience with open source tools, analyze legal and ethical considerations, and evaluate enterprise adoption strategies. By the end of the course, learners should be able to contribute effectively to open source projects and apply open technologies in real-world environments.

Expected Learning outcomes

By the end of the course, students will be able to:

1. Define and explain at least five core concepts of open source systems, including licensing, governance, and community development.
2. Differentiate between at least three open source licenses and justify their appropriate use in given scenarios.
3. Install and configure a Linux-based operating system environment for development purposes.
4. Use Git to create, clone, branch, merge, and manage repositories in a collaborative workflow.
5. Contribute at least one documented change (code, documentation, or issue report) to an open source project.
6. Analyze governance structures of two open source communities and present findings in a written report.

7. Deploy a basic open source web application using a LAMP or similar stack.
8. Evaluate security risks in open source components using standard vulnerability assessment techniques.
9. Conduct a cost-benefit analysis comparing open source and proprietary solutions for a given case study.
10. Develop and document a simple open source project plan including roadmap and release cycle.
11. Demonstrate proper open source documentation practices using README and contribution guidelines.
12. Assess legal and ethical considerations in open source adoption within organizational contexts.
13. Collaborate effectively in a team using distributed development tools and communication platforms.
14. Present a functional prototype built using open source technologies to peers and instructors.
15. Critically evaluate emerging trends in open source systems and propose innovative applications in real-world contexts.

Examinations

- The University examinations regulations shall apply
- The University Common Rules and Regulations for undergraduate Examinations shall apply.
- Examinations shall be held at the end of the semester in which courses are taught
- Admission to Examination will depend on satisfactory attendance of the prescribed courses as per senate regulations.
- Continuous assessment tests and projects, account for 40% and, the final three-hour written examination will account for 60% of the final grade.

Course outline

Week 1: Introduction to Open Source Systems

Topics:

- Definition and evolution of Open Source
- History of the Open Source Movement
- Key terminology (FOSS, FLOSS, OSS)
- Benefits and challenges of open source systems

Learning Outcomes:
By the end of this week, students should be able to:

1. Define Open Source Systems and related concepts.
2. Differentiate between open source and proprietary software.

3. Explain the historical development of the open source movement.
4. Identify key milestones in open source history.
5. Discuss advantages and limitations of open source systems.

Week 2: Open Source Licenses and Legal Frameworks

Topics:

- Copyright and intellectual property basics
- Permissive vs. copyleft licenses
- Popular licenses (GPL, MIT, Apache)
- License compatibility issues

Learning Outcomes:
Students will be able to:

1. Explain intellectual property rights in software.
2. Distinguish between permissive and copyleft licenses.
3. Compare common open source licenses.
4. Analyze license compatibility in multi-component systems.
5. Evaluate legal risks in open source adoption.

Week 3: The Open Source Ecosystem

Topics:

- Open source communities
- Governance models
- Foundations (e.g., Apache, Linux Foundation)
- Community contribution models

Learning Outcomes:
Students will be able to:

1. Describe the structure of open source communities.
2. Explain governance models in open source projects.
3. Analyze how foundations support OSS.
4. Identify roles within OSS communities.
5. Assess factors that influence project sustainability.

Week 4: Open Source Development Tools

Topics:

- Version control systems (Git)
- GitHub/GitLab workflows

- Issue tracking systems
- Continuous integration basics

Learning Outcomes:
Students will be able to:

1. Explain distributed version control concepts.
2. Use Git for basic repository management.
3. Demonstrate collaboration using pull requests.
4. Manage issues and project boards.
5. Apply basic CI/CD principles in open source projects.

Week 5: Open Source Operating Systems

Topics:

- Linux architecture
- Distributions (Ubuntu, Fedora)
- Kernel vs user space
- Shell and command line basics

Learning Outcomes:
Students will be able to:

1. Describe the architecture of Linux systems.
2. Compare different Linux distributions.
3. Navigate and manage files using the command line.
4. Explain the role of the Linux kernel.
5. Install and configure a Linux distribution.

Week 6: Open Source Databases and Middleware

Topics:

- MySQL, PostgreSQL
- Application servers
- LAMP/LEMP stack
- Open source middleware

Learning Outcomes:
Students will be able to:

1. Differentiate relational open source databases.
2. Install and configure an open source DBMS.
3. Explain middleware functions.
4. Deploy a simple web application using LAMP/LEMP.

5. Evaluate performance considerations.

Week 7: Open Source in Enterprise Systems

Topics:

- ERP and CRM systems
- Case studies (Odoo, ERPNext)
- Adoption strategies
- Cost-benefit analysis

Learning Outcomes:
Students will be able to:

1. Identify open source enterprise applications.
2. Analyze the benefits of OSS in organizations.
3. Compare proprietary vs OSS ERP solutions.
4. Conduct basic cost-benefit analysis.
5. Recommend OSS solutions for SMEs.

Week 8: Open Source in Cloud and Virtualization

Topics:

- OpenStack
- Docker and Kubernetes
- Virtual machines
- Cloud deployment models

Learning Outcomes:
Students will be able to:

1. Explain virtualization concepts.
2. Describe open source cloud platforms.
3. Deploy containerized applications.
4. Differentiate IaaS, PaaS, and SaaS models.
5. Assess scalability benefits of cloud OSS.

Week 9: Security in Open Source Systems

Topics:

- Common vulnerabilities
- Secure coding practices
- Patch management
- Community security response

Learning Outcomes:
Students will be able to:

1. Identify common open source security risks.
2. Explain vulnerability disclosure processes.
3. Apply secure coding practices.
4. Evaluate security update mechanisms.
5. Assess risks in third-party dependencies.

Week 10: Open Source and Emerging Technologies

Topics:

- AI/ML open source tools
- Blockchain platforms
- IoT frameworks
- Big Data tools (Hadoop)

Learning Outcomes:
Students will be able to:

1. Identify open source tools in AI and data science.
2. Explain OSS use in blockchain systems.
3. Describe open source IoT platforms.
4. Analyze OSS role in big data analytics.
5. Evaluate innovation trends driven by OSS.

Week 11: Open Source Project Management

Topics:

- Agile in open source
- Roadmaps and release cycles
- Documentation practices
- Community communication

Learning Outcomes:
Students will be able to:

1. Describe agile methodologies in OSS.
2. Create a simple open source roadmap.
3. Develop project documentation.
4. Apply collaboration tools effectively.
5. Evaluate contributor engagement strategies.

Week 12: Contributing to Open Source Projects

Topics:

- Forking and pull requests
- Code reviews
- Writing documentation
- Bug reporting

Learning Outcomes:
Students will be able to:

1. Fork and clone repositories.
2. Submit meaningful pull requests.
3. Participate in code reviews.
4. Write clear documentation.
5. Contribute to community discussions.

Week 13: Open Source Policy, Ethics and Sustainability

Topics:

- Government OSS policies
- Ethical considerations
- Digital sovereignty
- Sustainability models

Learning Outcomes:
Students will be able to:

1. Discuss public sector OSS adoption.
2. Evaluate ethical issues in OSS.
3. Explain digital sovereignty concepts.
4. Analyze sustainability challenges.
5. Propose governance improvements.

Week 14: Capstone Project Presentations and Course Review

Topics:

- Student project demonstrations
- Peer evaluation
- Reflection on learning
- Future trends in open source

Learning Outcomes:
Students will be able to:

1. Present an open source-based solution.
2. Demonstrate technical implementation.
3. Critically evaluate peer projects.
4. Reflect on open source career pathways.
5. Synthesize course concepts into practice.

Prepared by:	**Approved for issue**
……………………………..…	………………………………….
Dr. Gilbert Mugeni: Course Lecturer	HoD Department of Informatics and Innovation